Under the hood, DABs are collections of Databricks artifacts (including jobs, DLT pipelines, and Machine Learning models) and assets (e.g., notebooks, SQL queries, and dashboards). These bundles are managed through YAML templates created and maintained alongside source code. You can build DAB YAML files manually or use templates to automate. You can also build custom templates for more complex processing tasks.

Using DABs requires the Databricks CLI. We discussed installing the CLI in *Chapter 2* if you want to review it again. Also, DABs are fairly new and not incorporated into the projects. However, great resources are listed in Further reading that cover this new product feature in depth.

Rest API

Everything you can do in the UI can be accomplished via the API as well. The UI is great for exploring product features and building workflows for the first time. For example, we automated the process after building out our AutoML experiment in the UI. Additionally, we saw how secrets are completely contained in the API and not available via the UI. As we see in the next section, it is possible to deploy your models via API.

Deploying your model

Deploying a model can be done in many ways, depending on the use case and data availability. For example, deployment may look like packaging a model in a container and deploying at an endpoint or a model that runs daily in a production workflow to provide predictions into tables consumed by applications. Databricks has product features to pave the way to production for all inference types.

Model Inference

We've walked through the methods and tools that help you set up your model in production, and finally, you have a model ready for inference! But one key question you should consider as part of this process is how your model should be used. Do you need the results once a day? Is the model powering an application that requires real-time results? Your model's purpose will help you decide the type of deployment you need. You've seen the words "batch" and "streaming" a few times in this chapter already, so let's quickly define what those mean in the context of model inference.

- **Batch Inference**: Batch inference (also known as offline inference) refers to a job that generates predictions on a group (or "batch") of data all at once. Batch jobs are scheduled to run on a specified cadence, such as once a day. This type of inference is best when there are no/low-latency requirements and allows you to take advantage of scalable compute resources.
- **Streaming Inference**: Streaming inference (also known as online inference) refers to a job that generates predictions on data as it is streamed. This is possible in the Databricks platform via Delta Live Tables.